

Anthony Esposito

SOFTWARE ARCHITECT · APPLIED AI & AGENT SYSTEMS · MARKETING ANALYTICS

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Two careers, one method. For twenty-three years I turned household-level data into targeted, profitable marketing and ran operations in a regulated industry. **On May 1, 2026 I wrote my first real software** — a guitar tutor, because I wanted to learn to play — and I built it with AI, on purpose, from the first day. In the hundred days since, I have designed and built, as a sole engineer, **fourteen multi-tenant vertical SaaS operating systems, thirty-one consumer applications, and the agent runtime that operates them** — including a typed message bus, a validation layer that makes bluffed confidence arithmetically impossible, and a multi-agent organizational pattern documented in a working thesis. Everything listed here is live and enterable in a browser.

Layer 0 — Typed Message Bus PYTHON

Synchronous dispatch over a registry of named node handlers. Every message is typed on three axes — kind, direction and hop budget. Loop prevention is a TTL rather than a heuristic: the effective ceiling is `min(bus.max_hops, msg.max_hops)`, so a node may tighten its own budget but can never grant itself more runway. Hardened by adversarial review that found and closed three real defects, including a silent-loss path disguised as successful delivery.

The Output Contract HONESTY AS A SCHEMA

A frozen structure of stance, integer confidence 0–100, and two to three reasons, each tagged `data` or `assumption`. Conviction is derived, never claimed: if no reason is data-backed, the conclusion is forced to low-conviction regardless of the stated number. Oversize payloads are rejected rather than truncated, and a null payload raises *"silence is not allowed."*

Multi-Tenant White-Label Factory 14 VERTICALS LIVE

A tenant-agnostic engine in which one configuration file is the only swap point — brand, skin, vocabulary, rooms, roles and AI persona. Each tenant deploys as its own project with its own environment, so isolation happens at the platform boundary rather than through row-level filtering. Every API is environment-gated with graceful degradation, so a cold clone renders and demonstrates with zero credentials configured.

The Spine & Rooms Model DOMAIN MODELING

Each vertical is modeled around a single core domain object — a project, a listing, a show, a membership, a person — and every module page is a projection over that one record rather than a separate silo. Cross-room consistency becomes structural instead of synchronized, and adding a room stays additive and cheap.

Runtime Truth Enforcement AGENT RECEIPTS

A server-side guard that forces a real tool call and emits a receipt, so an agent cannot narrate work it did not perform. Paired with a resumable worker that recovers tasks stranded by function timeouts, and a cron heartbeat that lets seats pull from and push results to a shared task board.

Native Realtime Communication OWN WEBRTC

Video conferencing built on own signaling rather than a third-party SDK — one-to-one and group video, persistent channels, translated messaging, asynchronous video, and guest links joinable from a plain browser with no account and no per-seat license.

Engineering Governance 16 ARTICLES

A written constitution covering approval gates, credential fences, honesty rules (*"blank beats confident-wrong"*; every figure tagged live-data, estimate or assumption), a live-verification mandate that treats reporting a stale claim as fabrication, and a session-package rule requiring architecture, code, brief and handoff for every unit of work.

Direction Invariants ORG CHART AS TYPES

Only a packaged conclusion or a flagged terminal notice may travel upward through the tree; a question moving up is rejected by the transport itself. Escalation policy is compiled into the message type system instead of being requested in a prompt. Every failure mode — hops exceeded, direction violation, contract violation, unregistered target — emits a distinct terminal notice back to origin with its own counter.

The Triad Pattern ANTI-GROUPTHINK

Two specialist lenses — one quantitative, one interpretive — deliberately non-overlapping and structurally sealed from one another so they cannot converge socially. A single pacemaker reconciles them, maps their agreement to a confidence figure, applies a release gate, and is the only voice out. Any lens reading unfavorable holds a veto, and a domain-specific honesty cap prevents a confident recommendation on modeled data.

Programmatic Provisioning TEXT TO BUSINESS

A build pipeline that creates a repository, creates a hosting project and starts a fresh branded product end to end against the GitHub and Vercel APIs, driven from a visual front end where a vertical is chosen, branded and generated. SaaS tenant provisioning, operated by one person.

Safety Enforced by Absence LEARNER PRODUCTS

Guarantees implemented as missing capability rather than instruction. There is no timer code, so a child cannot be raced; no error-red token, so failure cannot be rendered; no schema field where agent-authored prose could ride, plus an n-gram fence rejecting content words the child did not write. A rule expressible as configuration is one write from repeal; a rule that is structurally unrepresentable is not.

Skins Engine ACCESSIBILITY GUARD

Per-user themes selected at sign-in and applied without flash on page transition, with a runtime contrast auto-fixer that computes contrast ratios and forces readable text on any light-on-light control — touching only broken pairs and never intentional color.

Self-Hosted Mail & Groupware LINUX · \$0/MONTH

Provisioned an Ubuntu server on free-tier cloud infrastructure and stood up a complete mail system: Postfix receiving on 25, Dovecot IMAP over TLS on 993, authenticated submission on 587, and Radicale CalDAV/CardDAV on 5232, fronted by a Let's Encrypt certificate with automatic renewal. **SPF, DKIM at 2048 bits, and DMARC all published and passing.** Cut the MX record over from Google Workspace, retiring the subscription — five mailboxes plus catch-all forwarding across two domains, with legacy addresses preserved indefinitely. Worked around the provider's blanket outbound port-25 block with an authenticated SMTP relay, and added two gigabytes of swap to keep the services stable on a 954 MB instance. The console provisioning form proved unusable, so the whole box was built through the cloud CLI. Delivery confirmed in the mail log rather than assumed.

02 PRODUCTS SHIPPED — BUSINESS PLATFORM

Fourteen multi-tenant white-label operating systems, each live and enterable as a gated showroom running the real software on seeded data. Monthly list prices \$99–\$650.

PRODUCT	VERTICAL	DOMAIN MODELED
Abode OS	Real estate	Listings, deal spine, transaction coordination
Buttress OS	Architecture	Drawings, phases, submittals, the books that compute off the same record
Truss OS	Engineering	Pursuits and go/no-go, WBS, field RFIs, calcs and standards, PE-seal gate, permits and AHJ, multipliers — with a discipline module switch for civil, mechanical and software
Smiley OS	Dental practices	Chair time, treatment plans, the practice money engine
Stay@Home OS	Home care agencies	Caregiver scheduling, visit records, care plans
Toolbelt OS	HVAC, plumbing, electrical	Dispatch switchboard, web booking, trucks and jobs
4barrel OS	Used-car dealers	Inventory, reconditioning, desking
Musical OS	Live theater	Box office, seat authority, production spine, program
8mm OS	Film & production	Slate, crew, schedule, settlement
Amphitheater OS	Concerts & venues	Holds and confirms, settlement, day-of-show
Moments OS	Photo & video	Sessions, galleries, delivery, a solo-class price floor
Sleeves OS	Tattoo studios	Artists, deposits, booking, consent
Targeted OS	Marketing agencies	Campaigns, audiences, creative rounds, reporting
LilNinja OS	Kids gyms & programs	Memberships, sessions, waivers, family accounts

03 PRODUCTS SHIPPED — CONSUMER

Thirty-one applications across nine shelves, most free to try, all live.

Music & making	ESPO Music — five instrument coaches (guitar, piano, harmonica, ukulele, bass) from one shared engine with Karplus–Strong synthesis. ESPO Fret, the guitar coach, was the first software I ever wrote — May 2026, because I wanted to learn to play; now a shipping product with subscription billing · ESPOfunkmaster, a browser studio with drums, keys, looper, mic, mixer and record cutting · ESPO Cutlabs, a video editor · AE Cre8, eight machines that make things
Learning	ESPO Learning (handwriting, reading, math, composition — no fail states, COPPA-clean) · ESPOhystory · ESPOvocab · ESPOwords · ESPOdraw
Mind & mood	Neuro Divulge regulation tools · The Thread · ESPOtendency · ESPOregulator · ESPOfocus
Money & fine print	ESPO Genius (plain-English paperwork) · The Narcs — The Narc and MarketNarc, a fundamentals-and-technicals analysis triad · ESPOsign
Journals	ESPOvineyard · ESPOstogie · ESPOwhiskey · ESPObarista — on-device, no ads, no marketplace
Travel & sport	ESPOtrek · ESPOnatlparks (offline passport, 63 parks) · ESPOracket
Family & other	ESPOnest · ESPO Drama · Revolver OS · Sparkle Salon

Functional-medicine telehealth practice — premium build

2026

Installable progressive web app with a native tab bar; native WebRTC video visits; translated patient messaging; asynchronous video check-ins between appointments; AI-assisted clinical note drafting; a talking in-app assistant with voice out and speech in; separate staff and patient themes; and a written access model covering data ownership, least-privilege roles, logged administrative entry, an accounting fence, and a revocation path that cuts the vendor's access without touching anything the client owns.

Regional nonprofit theater — full operating system, pro bono

2026

In-house ticketing with server-authoritative seat holds replacing a third-party dependency; box office and point of sale; CRM with cross-domain person joins uniting roles, crews and classes on one record; patron lifetime-value history; nonprofit books to 990 standard with an audit-safe void path; role-aware navigation across eleven staff and volunteer roles; records management; a show library and scene shop; a program builder; marketing and newsletter tooling; and an in-hub AI creative director. A companion advisory triad monitors box-office health on a clear / watch / intervene scale.

Engineering firm · youth sports organization — vertical builds

2026

Phase-native project spines from pursuit through invoice, field logs, permitting workflow, books with utilization multipliers, and departmental agent routing.

05 AEPSYCHOGRAPH — HOUSEHOLD SEGMENTATION ENGINE

The premise. Rebuild PRIZM-class household segmentation on free public government data, minus the license layer. Commercial comparables list between \$1,645 per year and \$1,500 per month.

The model. Forty-eight segments across four urbanization tiers, four affluence tiers and three lifestages; fourteen activities; eight psychographic types; indices computed as a segment's rate over the national average, times one hundred, clamped to a 20–320 band.

The sources. US Census American Community Survey five-year tables for income, households with own children, and 65-and-over; Bureau of Economic Analysis state personal consumption; Bureau of Labor Statistics Consumer Expenditure series, cached and refreshed on a thirty-day cycle.

The honesty constraint, enforced in code. Who a household is comes from real Census data; what they are likely to do is modeled — and the two are never allowed to blur. Without live Census data the pacemaker is hard-capped below the targeting threshold, making a confident targeting recommendation on modeled data **arithmetically impossible rather than merely discouraged**. Classifiers are required to return "unclassified" and "unassessable" rather than force a label.

Founder & Principal Engineer — Accelerated Experiences LLC · Post Falls, ID

MAY 2026 — PRESENT

First line of code written May 1, 2026. LLC formed at the end of that month. Everything in sections 01 through 05 was built in the hundred days that followed.

- Sole architect, engineer, designer and operator of everything catalogued in sections 01 through 05.
- Owned the commercial side end to end: pricing architecture across the fleet, a single authoritative price record with per-product locks, packaging and tiering, and a deliberate no-per-seat-pricing promise enforced as a product constraint.
- **Recruited and equipped two commission sales representatives** across six states with territory playbooks, objection handling and a 372-record qualified lead database.
- Built the media production stack that markets the work: procedural video render pipelines, a voice engine, and a browser video editor — used to produce product films for every vertical.

Homeschool Educator — North Idaho

2024 — 2026

Homeschooled four children, grades K–6, designing curriculum and building the educational tools that became the ESPO Learning product line — and the direct source of the no-fail-states-by-architecture safety rule.

MARKETING & OPERATIONS LEADERSHIP — SENIOR LIVING AND HOSPICE · 2010–2024

General Manager — Harvard Park Retirement Center (Holiday Retirement)	2016 — 2019
Marketing Director & Community Education — River City Hospice	2014 — 2016
Marketing Director — Avalon Care Center, North Spokane	2012 — 2013
Marketing Director — Auburn Crest Hospice, Post Falls	2010 — 2012

Fourteen years of P&L ownership, census and referral-network growth, community education programming, and compliance-bounded operations in regulated healthcare.

Marketing Analyst — Bay Area News Group, I-80 Network · Vallejo, CA

2008 — 2010

Launched an analytics department spanning three daily newspapers — the Vallejo *Times-Herald*, the Woodland *Daily Democrat* and the Vacaville *Reporter* — including requirements, data model, custom system build and rollout across all three properties.

Market Researcher & Database Marketer — Sales Farm · California

2006 — 2008

Market research and PRIZM segmentation analysis for a startup agency's client roster.

Marketing Analyst — The Press Democrat, New York Times Media Group · Santa Rosa, CA

2003 — 2006

- Promoted from temp staffer to Marketing Analyst in two years; **built the company's direct-mail system** from the data model up.
- Led analytics across the daily and its sister publications — *Savor*, *Wine Country*, the Petaluma and Santa Rosa magazines, and the associated web properties.
- Applied **household-level PRIZM / Claritas segmentation** to editorial, circulation and advertiser targeting; supported major retail accounts including Macy's, Gottschalks and Lowe's.
- **Ran print production on a physical board** — a card for every job moving through designer, printer, mailhouse, list procurement, drop date and in-home date — and built an early estimating, production and accounting system so the gross margin set at the estimate was the figure the books were held to. **That is the spine model described in section 01, arrived at two decades before I could express it in software.**

Administrative Assistant to the HR Director — City of Santa Rosa, CA · summer position during college

SUMMER

The Founder Triad: How One Human and Two Instances of the Same AI Built a Production SystemJUN 2026

A thesis on the working method behind the platform. Argues that a single operator holding the pacemaker seat between a design instance and a build instance can stand up a product family with production discipline, provided shared knowledge lives in documents rather than memory, safety promises are enforced by architecture rather than instruction, and one seat is never delegated. States precisely which seat that is and why: the errors that mattered were errors *inside the canon* — the one place referees and tests cannot look. Closes with a section titled **Honest Limits** enumerating everything the system had not yet earned the right to claim.

The AETriad Training Playbook v1.0JUN 2026

The reusable method for building and tuning an agent triad, framed around five levers — Role, Knowledge, Tools, Output Contract, Feedback Loop — with the claim that only the fifth constitutes real learning, and a fully worked expert-review → diagnose → patch → route-evidence → verify episode as the reference case.

The AE Constitution2026, LIVING

Sixteen articles of engineering governance: authority and approval gates, credential fences, truth and honesty rules with a formal output contract, live-verification mandates, a quality bar, brand law, delivery and session-package requirements, shipping cadence, memory and continuity, and learner safety.

LANGUAGES	Python 3 · Node.js (ESM) · JavaScript · HTML · CSS · SVG
AI & AGENTS	Anthropic Claude · DeepSeek · ElevenLabs · prompt and role architecture · evaluation harnesses and golden-data referees · adversarial red-team corpora · confidence gating and derived-conviction models · tool-use and output-contract design · multi-agent orchestration and routing · retrieval over canonical documents
PLATFORM	Vercel serverless functions, multi-project tenant isolation, cron scheduling, redirects · GitHub · key-value and blob storage · environment-gated graceful degradation · single sign-on and role-based access control
INFRASTRUCTURE	Ubuntu 24.04 on Oracle Cloud · CLI-based instance provisioning · Postfix · Dovecot · Radicale · Let's Encrypt and TLS certificate automation · SPF, DKIM and DMARC · DNS, MX and zone administration · SSH key management · swap and resource tuning on constrained hosts
REALTIME	WebRTC with own signaling · Web Speech API · progressive web apps and service workers · offline-first storage
DATA	US Census ACS · BLS Consumer Expenditure · BEA SAPCE · FRED · segmentation and index modeling · PRIZM / Claritas methodology · FileMaker
MEDIA	Python with PIL and NumPy for procedural rendering · FFmpeg and FFprobe · CairoSVG · ReportLab · WeasyPrint · voice cloning and audio-video muxing · on-device OCR · Karplus–Strong synthesis
COMMERCE	Stripe · Google Play publishing · Google Search Console, sitemaps and technical SEO · pricing and packaging architecture
VERIFICATION	Python conformance suites · property-based tests · browser suites · syntax gates on every deployed module · live-URL verification as a release requirement

University of Idaho — B.S. Media Communications, Advertising & Public Relations	1999 — 2003
Chairman's Award — The Press Democrat, New York Times Media Group	—
Presenter, Claritas national conference — San Diego, CA	—
Mentored by Bill Truett in household segmentation methodology	—